

COMP3242/6242: Deep Learning

Week 12 — Review

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Deep Learning

This course has taken you from the fundamentals to some of the latest and most advanced topics in deep learning:

1. Introduction and Propaedeutics
2. Linear Classifiers and Mult-layer Perceptrons
3. Automatic Differentiation and Back-propagation
4. Image Classification, Object Detection and Convolutional Neural Networks
5. Image Segmentation and Deconvolutions
6. Sequence Models and Back-propagation through Time
7. Developing and Debugging Algorithms
8. Attention and Transformers
9. Contrastive Learning
10. Deep Reinforcement Learning
11. Generative AI

but we only just scratched the surface...

Ingredients

- ▶ data and labels (supervision)
- ▶ representation (datastructures)
- ▶ model/network architectures
- ▶ objectives (losses and priors)
- ▶ learning algorithms
- ▶ evaluation metrics

plus:

- ▶ tools
- ▶ communities
- ▶ infrastructure

Data and Labels

- ▶ **supervised:** $\mathcal{D} = \{x^{(i)}, y^{(i)}\} \subset \mathcal{X} \times \mathcal{Y}$
 - ▶ classification, $\mathcal{Y}$ is a discrete set of labels
 - ▶ regression, $\mathcal{Y}$ is equivalent to $\mathbb{R}^n$
- ▶ **unsupervised:** no labels, $\mathcal{D} = \{x^{(i)}\}$
- ▶ **self-supervised:** derive $y^{(i)}$ from $x^{(i)}$
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- ▶ balanced versus unbalanced

Data and Labels

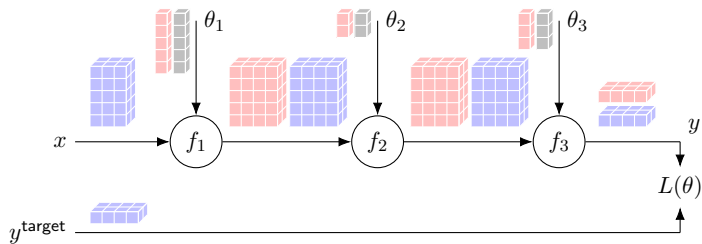
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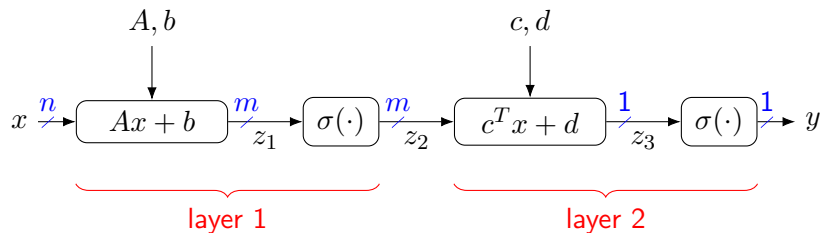
- ▶ all data is noisy — quality versus quantity
- ▶ many learning algorithms can handle some noise
- ▶ evaluation (hence conclusions) may be affected by noise

Representations

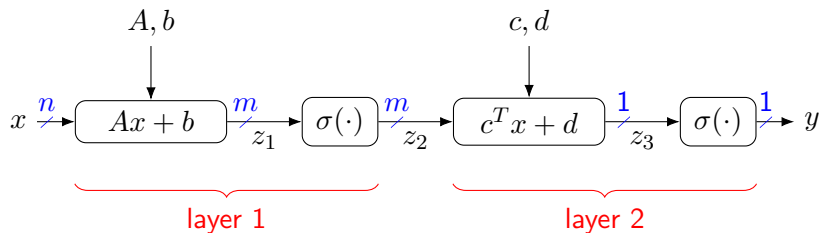
- ▶ deep learning models operate on numerical tensors
 - ▶ features vectors are already (1-dimensional) tensors
 - ▶ images are already tensors, $3 \times H \times W$
 - ▶ labels (discrete symbols) are mapped to $\{1, \dots, K\}$ or one-hot vectors
 - ▶ text (symbolic sequences) is tokenized and then tokens are encoded as vectors
- ▶ models process data in mini-batches



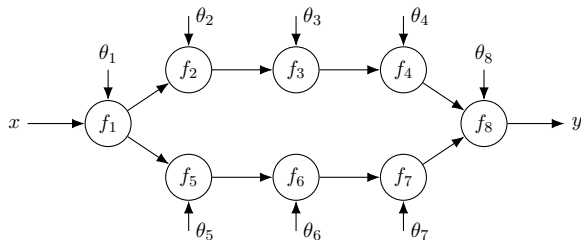
Models and Architectures



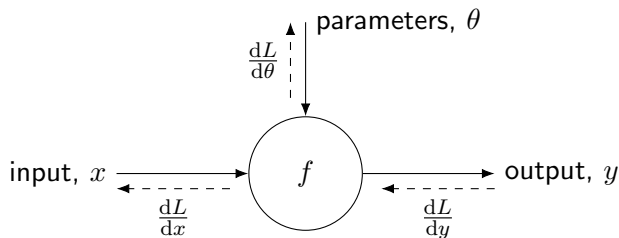
Models and Architectures



- ▶ linear classifiers
- ▶ multi-layer perceptrons
- ▶ convolutional neural networks
- ▶ recurrent neural networks
- ▶ transformers



Deep Learning Node (aka Layer)



► **Forward pass:** compute output y as a function of the input x (and model parameters θ).

► **Backward pass:** compute the derivative of the loss with respect to the input x (and model parameters θ) given the derivative of the loss with respect to the output y .

Objectives (Losses and Priors)

- ▶ **supervised:** cross-entropy loss, mean-square-error loss
 - ▶ **object detection:** classification loss and bounding box loss
 - ▶ **self-supervised:** reconstruction loss, contrastive losses
 - ▶ **language models:** next-token prediction (cross-entropy loss)
 - ▶ **reinforcement learning:** (discounted) sum of future rewards
 - ▶ **generative AI:** evidence lower bound (ELBO)
-
- ▶ ℓ_2 -loss on parameters is weight decay

$$\theta \leftarrow \theta - \eta \nabla L(\theta)$$

- ▶ stochastic gradient descent and variants (e.g., AdamW)
- ▶ initialization is important; mini-batch sampling is important
- ▶ back-propagation via automatic differentiation

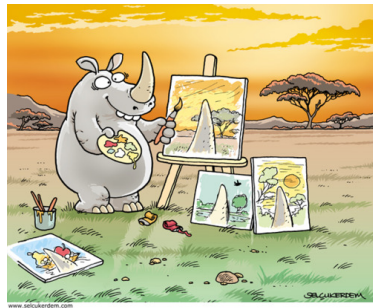
Evaluation Metrics

it's essential to evaluate on held-out test data:

- ▶ the loss is only a surrogate — we often can't directly optimize the thing we really care about
- ▶ training data is only a sample of the population

this is hard for models trained on massive amounts of data and for semantic tasks

- ▶ don't just run experiments for evaluation
 - ▶ also run diagnostics
- ▶ visualize results



Beyond This Course — Some Topics We Didn't Have Time to Cover

...but we only just scratched the surface

- ▶ more exotic architecture variants
 - ▶ graph neural networks, mixture-of-experts, state space models, implicit neural representations, coordinate networks, spiking neural networks, neural radiance fields
- ▶ advanced learning algorithms
 - ▶ LoRA, Muon, distributed learning
 - ▶ theory and convergence rates, scaling laws
- ▶ deep declarative networks, neural ODEs, and physics informed networks
- ▶ world models, vision-language-action models
- ▶ neural architecture search, distillation, quantization and pruning
- ▶ systems: GPU and CUDA fundamentals
- ▶ ethics and safety, alignment, interpretability

ANU Intelligent Systems Courses

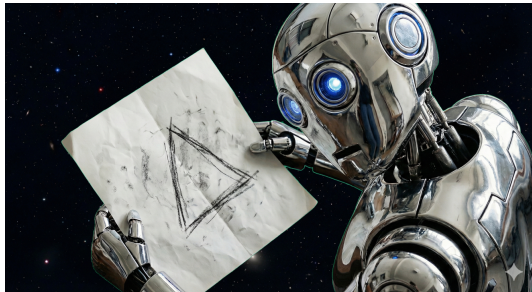
- ▶ COMP2610 Information Theory
- ▶ COMP3620 Artificial Intelligence
- ▶ COMP3670 Introduction to Machine Learning
- ▶ COMP4528 Computer Vision
- ▶ COMP4620 Advanced Topics in Artificial Intelligence
- ▶ COMP4650 Document Analysis
- ▶ COMP4670 Statistical Machine Learning
- ▶ COMP4680 Advanced Topics in Machine Learning
- ▶ COMP4691 Optimisation
- ▶ Projects

Magic

- ▶ AI systems built on DL can do remarkable things
 - ▶ we don't yet fully understand what and why
- ▶ remember under the hood they are nothing more than:
 - ▶ composed simple mathematical operations
 - ▶ clever engineering
 - ▶ trained on lots and lots of data



"competence without comprehension"



Last Word

- ▶ thank you for being great students
- ▶ good luck with your projects and exam, and future studies
- ▶ this is the first time we've run this course, please provide feedback